Accounting for Love of Variety in Consumer Demand Estimation

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1 Introduction

Standard discrete choice models in industrial organizational economics assume consumers are static agents who make independent choices each period. In reality, this is often not the case. Consumers may value products differently in one period than the last. In the realm of non-durable goods, this could be due to a habit stock the consumer has built through past consumption. Very real factors like boredom, preferences for novelty, or brand loyalty may attribute to time-variant preferences. However, including a channel through which past choices can shape current marginal utility, perhaps we can get better estimates of consumer preferences, allowing for improved measures of consumer welfare.

One explanation for consumers' preferences over time is love of variety. Love of variety research largely originates from marketing researchers, explaining why demand estimation would be more important. This is useful for things understanding the effect of advertising via display positioning or how a feature advertisement affects demand. Using Nielsen data on single-serving yogurt cups, I add to the literature on love of variety in consumer demand by including supply side effects. Marketers, to date, have largely focused on the demand side of the equation, leaving a gap in how firms internalize consumer preferences for variety over time. Further, reseearch which has looked into this phenomenon is theoretical. So, taking an empirical lens to the problem is a novel contribution. Single-serving yogurt is an attractive product space due to its relative lack of storability, its large flavor variety, and frequent purchase rate. Adding a supply side allows me to investigate how firms can price-discriminate or target their advertising through discounts based on consumer preferences for variety.

I estimate a discrete choice model of consumer demand, including a term which captures preferences for variety. The model uses purchase data in Atlanta, Chicago, Denver, and San Diego, whose variation in pricing I use for my control function. My key finding is that consumers exhibit a distaste for variety. This disutility is statistically significant and requires \$2.20 in compensation to switch flavors. Using this finding, my supply side counterfactual will examine how firms adjust pricing in response to this, attempting to induce switching to grow market share. I also find that, relative to other flavors, consumers prefer berry and plain yogurt, with berry being statistically significant.

The rest of the paper is organized as follows: section two discusses the related literature and section three describes the data used and some descriptive statistics. Section four showcases the model and environment and section five presents results. Section six goes into counterfactual analysis, while section 7 concludes.

2 Related Literature

The variety-seeking and state dependence literature has been mainly focused on understanding the dynamics of consumer preferences over time. When looking at earlier papers in this literature, McAlister [1982] proposes a model that accounts for consumers being attracted to different attraibutes at different times. Givon [1984] models how consumers vary over brands rather than attributes specifically, investigating how this impacts the choice set. Lattin [1987], building upon this, models the balance between current consumption and the lingering utility from past utility. Erdem and Keane [1996] then adds a learning channel to the model, allowing for experience and advertising to influence switching dynamics. Chintagunta [1999] is the most closely related paper, as he develops a model of variety-seeking behavior which is both tractible and flexible, and uses Nielsen data to estimate the model. However, this paper only looks at demand, leaving me a niche to explore the interplay between variety-seeking and strategic advertising. Dubé [2004] progresses Chintagunta's work, showing how consumers seek variety in quantities purchased. Using the soda market, Dube is able to model purchase occasions at a very granular level. I take attributes of both of their approaches to estimate the demand side.

The literature on personalized pricing is mostly theoretical outside of digital markets. Grossman and Shapiro [1984] creates a model to find the optimal level of advertising to heterogeneous consumers in a differentiated product market. Karle and Reisinger looks at the effect of imperfect targeting of consumers due to data privacy laws in digital markets, finding that the level of precision in targeting is massively important for the firm's advertising strategy. Moorthy and Shahrokhi Tehrani [2023] presents the theoretical effect of targeted advertising on the Hotelling model, showing that product differentiation is replaced by infomational differentiation. Finally, Dub'e and Misra [2023] uses a randomized control trial (RCT) to understand how price personalization impacts firm profits and consumer welfare. My contribution is bringing these theoretical papers, to an empirical realm. Further, I will be able to externally validate the RCT results from Dub'e and Misra [2023].

3 Data

3.1 Data

My data comes from Nielsen's HomeScan and MarketScan data. These two main datasets give me access to household purchase decisions as well as product characteristics. From HomeScan, I can access household characteristics, purchase histories by shopping trip, and information on purchase characteristics like if a discount was used or the total price paid. From MarketScan, I get product characteristics like price or if the product was in a featured position on the shelf. I limit the sample to Atlanta, Chicago, Denver, and San Diego in 2014. These markets were chosen since they are major cities in different parts of the country. I only include single-person households to avoid any contamination of shopping for multiple people like children or spouses. This ensures I am picking up true preferences for variety rather than buying for multiple people. These trips are encoded with unique identifiers, thus allowing me to look at per-trip, and within-trip, decisions. Agents are under observation for eight weeks and asked to record each shopping trip. Stores are restricted to be valid Nielsen stores, meaning they appear in the reatil code data. This is necessary to observe product prices.

I am looking at yogurt. Yogurt was chosen for a host of reasons. First, it is a good that is not storable indefinitely, being good for approximately one to three weeks. Yogurt also has a large menu of flavors within the category, allowing consumers to switch easily along this dimension. Similarly, yogurt has many types like sugary, Greek, Skyr, low-fat, and many more. This presents potential variation to exploit as well.

I classify flavors into three categories: berry, plain, and other. While these are not taking full advantage of the granularity in flavors available, the classifications capture the two largest categories, berry and plain. Berry and plain will then be compared to other flavors, which serve as the baseline category. I define the outside option as a trip in which yogurt was not ever purchased. Table 1 shows that the outside option is taken 80% of the time each trip. My sample has 1,269 agents who are seen for an average of about 18 trips. Of those 1,269, 715 made a yogurt purchase in 2014. If an agent was a yogurt purchaser, they purchased about 2.3 cups of yogurt per-trip. The mean income of my sample was 18,000. The sample is predominately white, with 75% of agents being White, 20% being Black, and 5% being either Asian or Hispanic.

Conditional on purchasing yogurt, 57% of households switch flavors at least once during the observation period. Approximately 50% of switching households did so using a coupon during purchase. It is impossible to know what product the coupon was used on, given multiple units are purchased each period. Yogurt buying households, on average, buy the same flavor for ten consecutive trips and switch three times.

Looking at figure 1, we can see the breakdown of switching behavior by flavor. For those who switched from berry yogurt to plain yogurt, they will stay on plain for 4.4 trips on average. Those who switch to plain tend to stay on plain longer for either case, staying an average of four periods if they switch from either flavor to plain. For those switching to other, they stay on that other flavor for 3.2 trips no matter the origin flavor. For berry, it appears that households are less stuck to it or use it as their switching flavor. That is, they break up the monotony of their primary flavor, plain or other, with a berry yogurt. This is interesting, since plain is more expensive on average. This implies the household's taste for plain yogurt outweighs their disutility of cost.

Finally, when looking at pricing across markets, we see interesting patterns. Firstly, table 2 shows that prices vary a lot between cities. In Chicago, for instance, berry is seen to have a mean price of \$0.84 per unit while plain has a mean unit price of \$1.22 per cup. This is a delta approximately twice the size of that between the two flavors in other markets. Looking at figure 2, it becomes clear that fluctuation in the overall mean price of yogurt cups fluctuates quite a bit throughout the year. In Atlanta and Denver, it seems they move largely in lockstep, while San Diego and Chicago are relatively more stable. San Diego and Chicago also have a different pattern than Atlanta and Denver do until around the middle of the year, when they start to move together. This variability across markets is what I exploit for the instrumental variable used in control function approach of Petrin and Train [2010].

4 Model

4.1 Consumer's Problem

The consumer's problem is the following. In each period t, the consumer i chooses a product j from the menu of available products J_t. The available products are $J_t = \{0, 1, 2, 3\}$ where $j = 0$ is the outside option, $j = 1$ is purchasing berry yogurt, $j = 2$ is purchasing plain yogurt, and $j = 3$ is purchasing other flavors. Consumers are assumed to be state-dependent, carrying previous choices forward. This history is summarized with $X^*_{i,t-1} \in J_t$, which is the modal flavor last selected by consumer i. Each product $j \in J_t$ is a bundle of $k \in K$ characteristics such that $X^k_{jt} \in \mathcal{X}_{jt}$. The consumer's utility from choosing product j at time t is given by:

$$U_{ijt} = \beta_0 + \sum_1^k \beta_i^k X_{jt}^k + \gamma_i \Xi_{ijt} + \alpha_i p_{jt} + \varepsilon_{ijt}$$

$$\begin{aligned} &\text{subject to} \\ &\qquad \Xi_{ijt} = |X_j t - X^*_{j,t-1}| \\ &\qquad \varepsilon_{ijt} \sim \text{T1EV}(0,1) \\ &\qquad\quad \text{for j} \in \{1,2,3\} \\ &\qquad\quad U_{i0t} \text{ normalized to } 0 \end{aligned}$$

In the utility function, there are the following components: β_0 is a constant, β_i^k is a consumer's utility from characteristic X^k_{jt}, γ_i is their preference for variety, Ξ_{ijt} is the difference between the current characteristic and their most recent choice, α_i is price sensitivity to p_{jt} and ε_{ijt} is the Gumbel shock which allows for a closed form logit. Choice probabilities take the following form:

$$P_{ijt} = \frac{\exp(\beta_0 + \sum_1^k \beta_i^k X_{jt}^k + \gamma_i \Xi_{ijt} + \alpha_i p_{jt} + \varepsilon_{ijt})}{1 + \sum_s \exp(\beta_0 + \sum_1^k \beta_i^k X_{st}^k + \gamma_i \Xi_{ist} + \alpha_i p_{st} + \varepsilon_{ist})}$$

Where we normalize the outside option, not purchasing yogurt, to be zero by construction.

5 Estimation

The estimation procedure uses standard maximum likelihood estimation. This means that there is no random coefficient estimation at this point, allowing me to identify the parameters of interest from the data alone.

Following Petrin and Train [2010], I use a control function to control for price endogeneity. I do a first stage instrumental variable estimation and then take residual from that estimation and insert it into the utility function. The first stage takes the form:

$$p_{jt} = \bar{p}_{j,-m(i),t} + \tau_t + \eta_{jt}$$

Where $\bar{p}_{j,-m(i),t}$ is the mean price from the three markets the household is not shopping in and τ_t is a time indicator. The residual corrects for the bias present in pricing, allowing us a consistent estimate of pricing.

I make an index of prices, instrumental variable residuals, and previous choices for each household's trips for each flavor. Then, I plug these values into the objective function represented by:

$$U_{ijt}^k = \beta_0 + \beta_i^k X_{jt}^k + \gamma_i \Xi_{ijt} + \alpha_i p_{jt} + \sigma \hat{\eta}_{jt}$$

Where β_0 is a constant, β_i^k for k $\in$ {berry, other, plain} are choice specific coefficients for choosing berry, other, or plain. $\gamma_i \Xi_{ijt}$ is the love of variety term, which is just capturing if you switched between periods. The price and control function term are unchanged from above. The probability of individual i choosing the observed product j in time t is given by:

$$\Pi_i (P_{ijt})^{y_{ijt}}$$

Where $y_{ijt} = 1$ when person i chosen j that period, and zero otherwise. The log-likelihood is then given by:

$$\text{LL}(\theta) = \sum_i^I \sum_j y_{ijt} \ln P_{ijt}$$

I then maximize the total log-likelihood to recover the parameters at the optimal.

6 Results

Results are shown in table 3. Here, you can see mixed findings. Firstly, I find that γ_i is statistically significant and negative with a value of -1.23. This finding points to consumers being less willing to switch flavors between trips. This estimate is at odds with what the data implies: more than half of houses are switchers, conditional on yogurt purchase, and houses switch 3 times on average if they do switch. This is likely a byproduct of not having random coefficients for utility from a given product. Further, I find a negative price coefficient α, but am unable to claim statistical significance. This is fixable with a better instrumental variable, something which is actively being considered. Similarly, the control function correction coefficient is not significant either. However, given σ is positive, we can infer that α is positively biased without this correction. With a better IV approach, the control function should also produce more impactful effects. I am looking into factors like distribution networks, other product categories, as well as different sources of geographic variation to serve as instruments.

When looking at the characteristic coefficients, only berry, β_i^b, is statistically significant. This is, again, an issue with lack of observations. As table 1 shows, berry has over 2,000 observed times being chosen. When looking at plain's coefficient, β_i^p, I do not find significance. This makes total sense upon seeing it is only chosen 412 times. The constant is negative, which is to be expected with a high outside option uptake rate. It is close to fringe significance, but is not either.

Finally, I calculated a willingness-to-pay for all β coefficients as well as the variety coefficient γ. These values are found in table 4. For the intercept, consumers value the outside option approximately \$1.50 more than yogurt flavored "other". For berry, it is valued about \$1.81 more than other flavors, while plain is valued about \$0.77 more. More interestingly, I find that consumers would need to be heavily compensated to switch. They would need approximately \$2.20 in incentives to switch from their normal choice of flavor. Judging by the average pricing of yogurt, this implies needing about two to three cups of yogurt given to them for free to switch.

Overall, my findings suggest consumers prefer to stick on flavors, experiencing disutility from switching. While a surprising result given the data facts, it must be noted that consumers do not just buy flavor. They buy for other characteristics that are not accounted for at this point in the research process. It will be interesting to see if this changes once I create a more holistic bundle of characteristics for the consumer to have preferences over. Similarly, the willingness-to-pay for variety is -2.20, meaning the consumer would need to be compensated \$2.20 to switch flavors. This opens a door for interesting counterfactual exercises in which the supply side is aware of the consumers willingness-to-pay and preferences, and thus try to influence them to switch.

7 Counterfactuals

Now, suppose we introduce firm side dynamics. Here, the firm acts strategically in a competitive market, taking the consumer's last observed action into account, and deciding whether or not to offer a promotion to encourage switching. This would lead to a dynamic pricing problem in which the firm is weighing the potential markup in the future against a lower price today. The firm will be able to oberve consumers' purchase histories, enabling them to first degree price discriminate. A firm would choose to discount a product j if the future market share it obtains offsets the lowered profits today. This also leads to analysis of competitive behavior between firms and how they decide to price depending on the behavior of other suppliers. Because the firm would not directly observe each consumer's type, I would be able to see the impact of targeting error on three dimensions: consumer welfare, competition in the market, and firm profits.

I also aim to look at a counterfactual involving product introduction. The firm would take its present approximation of the distribution over preferences and decide whether or not to introduce a product. Using a Hotelling model, I can test the impact of product introduction on switching behavior. With this, I am interested in consumer welfare, optimal product differentiation, and competitive effects.

8 Conclusion

In closing, this paper develops a dynamic discrete choice model of consumer demand that incorporates a love of variety and strategic pricing by firms. The model is estimated using scanner data, and counterfactual analyses are conducted to evaluate the effects of different market structures on consumer welfare and market outcomes. The paper contributes to the literature on love of variety, price-discrimination, and dynanmic consumer behavior, by providing empirical evidence on the effects of strategic pricing in a dynamic demand setting.

I find that consumers experience disutility from switching, needing to be compensated $2.20 to switch from one flavor to another. I find insignificant pricing effects, even with endogeneity intervention implying a need for a stronger instrumental variable. This simple version of the model suggests that there is significant runway for saturation with other product characteristics, random coefficients, and a richer sample.

The next iteration of this work will first augment the sample to include 2022-2024, as well as add markets. There is also room to add more characteristics like calories, grams of protein, grams of sugar, and type of yogurt. A brand effects pathway also makes sense, seeing if households are brand loyal or loyal to flavors. This decomposition of switching behavior would be very useful for the product introduction counterfactual especially. Lastly, the supply side will be an ongoing effort. With that being my primary contribution, it is obviously a priority and thus will be a major focus of forthcoming research.

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9 Tables and Figures

Table 1: Summary Statistics for Household Panel, Switching Dynamics, and Pricing

Statistic / Variable	Value	Unit / Description
Panel & Household Demographics		
Total Households in Sample	1,269	Households
Yogurt Purchasing Households	715	Households (56%)
Mean Trips per Household	17.65	Trips / Household
Mean Purchases per Shopping Trip	5.23	Units / Trip
Mean Household Income	35-40	Income Bracket ($,000's)
Median Household Income	40-45	Income Bracket ($,000's)
Outside Option Rate	80%	Non-purchase trip share
Average Age (Male):	25-29	Years Old
Average Age (Female):	40-44	Years Old
Racial Composition of Households		
White (Code 1)	955	Households (75%)
Black (Code 2)	254	Households (20%)
Asian (Code 3)	24	Households (2%)
Hispanic (Code 4)	36	Households (3%)
Switching & Purchasing Dynamics		
Ever-Switcher Households	57%	Conditional on yogurt purchase
Coupon Purchase Rate (All Purchases)	46%	Share of yogurt trips
Switching Attributable to Coupon	49%	Share of switch events
Mean Times Switching per Household	3	Switches / Household
Mean Consecutive Buys per Flavor	10.16	Continuous purchase trips
Yogurt Pricing		
Overall Mean Price	$0.84	Per unit price
Other Flavors Price	$0.76	Per unit price
Berry Price	$0.92	Per unit price
Plain Price	$1.16	Per unit price
Yogurt Choices		
Outside Option Chosen	8,102	Times Outside Option Chosen
Berry Chosen	2,108	Times Berry Yogurt Chosen
Other Chosen	1,865	Times Other Yogurt Flavors Chosen
Plain Chosen	412	Times Plain Yogurt Chosen

Figure 1

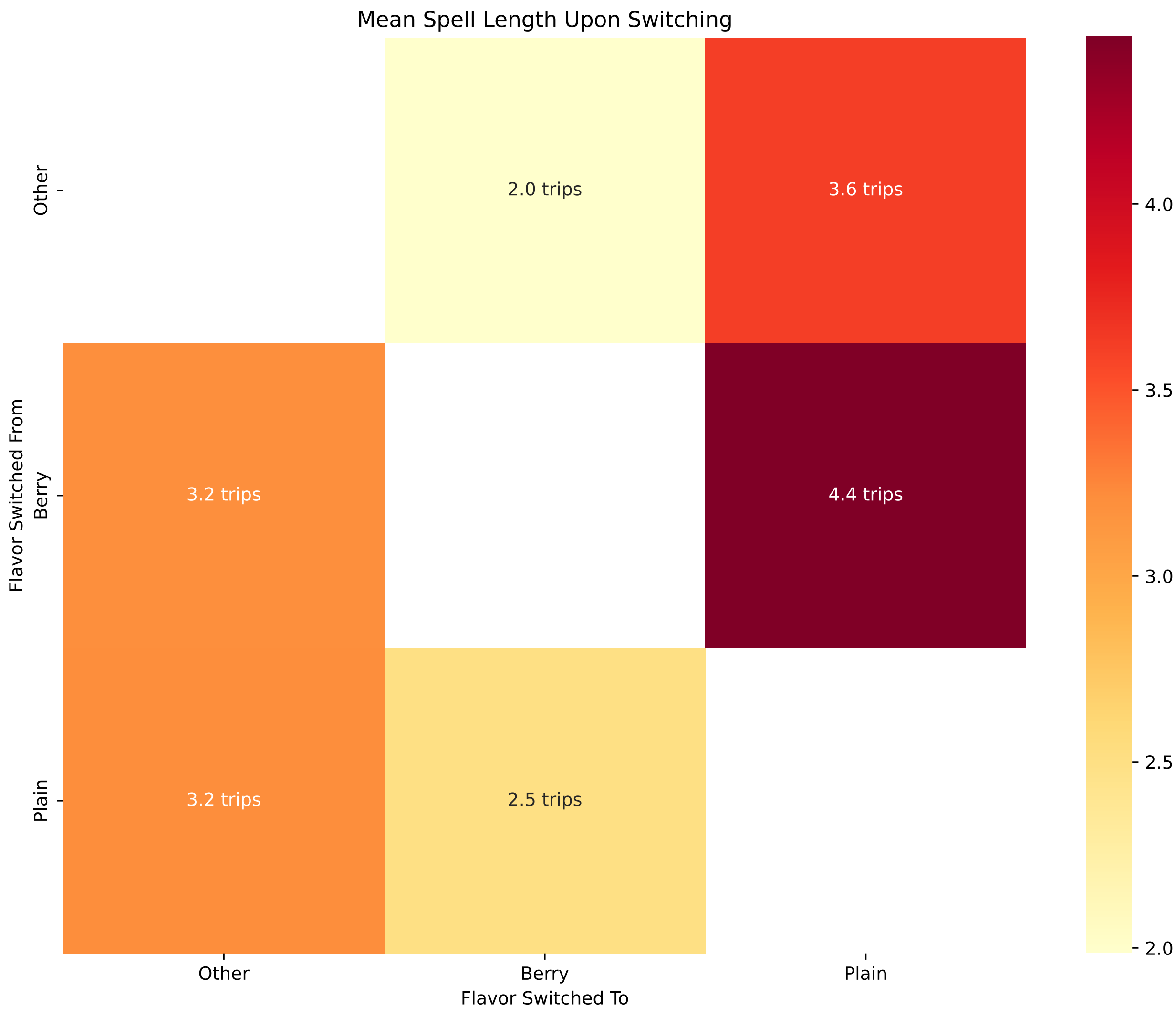
Mean Spell Length Upon Switching
2.0 trips
3.6 trips
3.2 trips
4.4 trips
3.2 trips
2.5 trips
Other
Berry
Plain
Flavor Switched From
Other
Berry
Plain
Flavor Switched To
4.0
3.5
3.0
2.5
2.0

Table 2: Mean Price by Market

Market Name	Mean Price (Berry)	Mean Price (Other)	Mean Price (Plain)
Atlanta	1.034	1.097	1.117
Chicago	0.842	1.038	1.224
Denver	0.999	1.101	1.118
San Diego	0.943	1.016	1.178

Figure 2

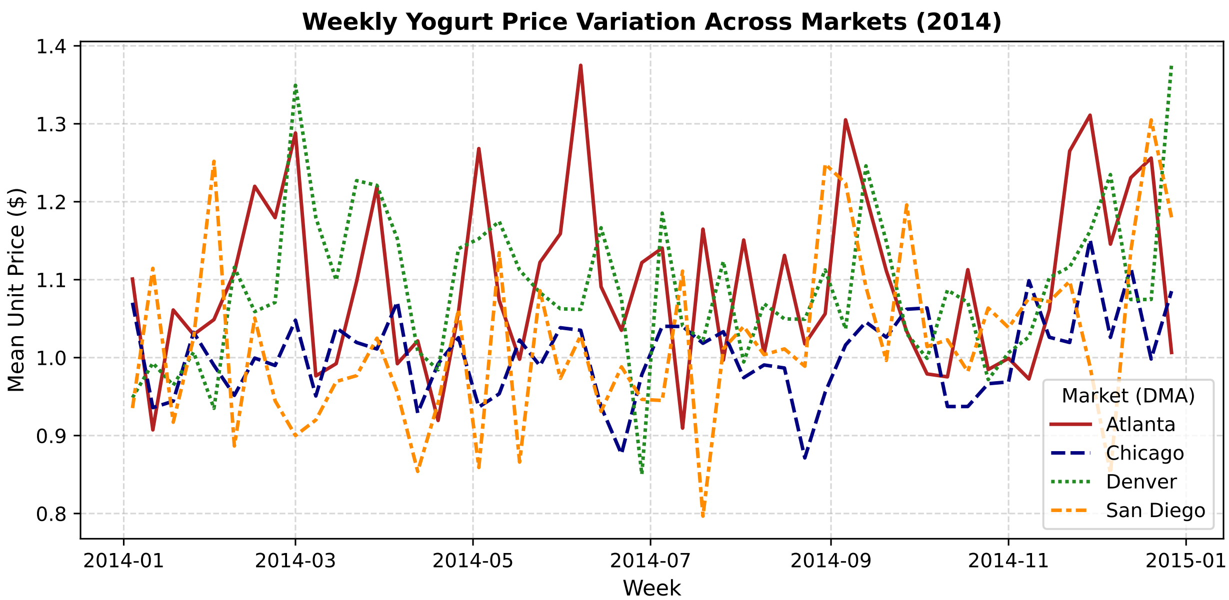

Table 3: Structural Estimation Results

Parameter	**Estimate**	**Std. Error**	**z-stat**	**p-value**
β_0 (Intercept)	-0.8402	0.5736	-1.465	0.14
β_{ber} (Berry)	1.0115	0.2485	4.070	0.00^{***}
β_{pl} (Plain)	0.4322	0.7620	0.567	0.57
γ (Habit/Variety)	-1.2333	0.5372	-2.296	0.02^{**}
α (Price)	-0.5603	0.5762	-0.973	0.33
σ (Control Func)	0.3831	2.7661	0.139	0.89

$^{***}p < 0.001$, $^{**}p < 0.05$

Table 4: Willingness-to-Pay (WTP) Estimates

Attribute / State	**WTP ($)**
Berry Flavor (β_{ber})	1.8053
Plain Flavor (β_{pl})	0.7714
Base Utility (β_0)	-1.5000
Habit/Variety (γ)	-2.2011

WTP calculated as $-$Parameter/α, where $\alpha = -0.5603$.